

ELVIS TANG

Embedded Systems | Firmware Engineering | C/C++

714-391-2768 | elvistang199@gmail.com

linkedin.com/in/elvis-np-tang | github.com/elvistang199 | elvistangprojects.com

Computer Engineering graduate with hands-on experience developing embedded C/C++ software for sensor-fusion and autonomous systems. Experienced with UART, SPI, I2C, Raspberry Pi, Arduino, real-time data acquisition, hardware/software integration, system testing, and technical documentation.

EXPERIENCE

Embedded Systems Project Lead - Lockheed Martin-Sponsored Capstone

Aug 2022 - May 2023

California State Polytechnic University, Pomona

Pomona, CA

- Developed and debugged embedded C/C++ software for an autonomous ground vehicle integrating LiDAR, stereo camera, ultrasonic sensors, IMU, and wheel encoders on Raspberry Pi and Arduino platforms.
- Implemented and tested UART, SPI, and I2C communication and Kalman-filter-based sensor fusion for real-time data acquisition and obstacle avoidance.
- Validated distance measurements within +/-10 cm on varied surfaces and demonstrated real-time object classification up to 50 cm during navigation trials.

Instructor

Nov 2023 - Sep 2024

Coding Minds Academy

Irvine, CA

- Taught Python, JavaScript, Git, game development, and introductory hardware concepts to classes of 10+ students.
- Guided student software and hardware projects through design, implementation, debugging, and demonstration.
- Developed project-based curricula connecting programming fundamentals with game development and real-world engineering tools.

Stand Lead

Aug 2024 - Present

Aramark

Anaheim, CA

- Patch and troubleshoot point-of-sale systems to keep registers operational during high-volume events.
- Coordinate daily concession operations serving 1,000+ guests per shift and resolve 20+ customer or operational issues.
- Manage inventory for 50+ items while maintaining food-safety, cash-handling, and workplace compliance procedures.

SELECTED PROJECTS

Sensor Fusion and Tracking for an Autonomous Ground Vehicle | C/C++, Raspberry Pi, Arduino, Kalman Filtering

- Integrated LiDAR, stereo camera, ultrasonic sensors, IMU, and wheel encoders into a multi-sensor perception and tracking architecture.
- Implemented UART, SPI, and I2C interfaces and fused real-time sensor data to support obstacle detection, localization, and autonomous navigation.
- Validated performance across varied surfaces with distance measurements within +/-10 cm and object classification up to 50 cm.

LucidGloves - VR Haptic Glove Technical Build Guide | ESP32, Haptics, 3D Printing, SteamVR

- Produced an illustrated implementation guide for an open-source VR haptic glove covering 3D-printed components, mechanical assembly, electronics, and wiring.
- Documented ESP32 firmware setup, SteamVR calibration, and end-to-end troubleshooting for repeatable hardware/software integration.
- Analyzed sensing, actuation, and haptic-feedback concepts that translate hand motion into responsive virtual-reality interaction.

Processor Architecture Performance Comparison | SilverBench, Benchmarking, Computer Architecture

- Ran repeatable SilverBench workloads to compare processor behavior and collect performance signals under a consistent test methodology.
- Interpreted results through computational throughput, workload behavior, and architecture-level tradeoffs.
- Presented test configuration, observations, limitations, and conclusions in a structured technical report.

EDUCATION & CERTIFICATIONS

B.S., Computer Engineering - California State Polytechnic University, Pomona

Certification: CompTIA Security+ (SY0-701)

TECHNICAL SKILLS

Firmware & Software: C/C++, Python, embedded debugging, real-time data acquisition, UART, SPI, I2C

Platforms & Hardware: Raspberry Pi, Arduino, ESP32, LiDAR, stereo camera, ultrasonic sensors, IMU, wheel encoders

Engineering Methods & Tools: sensor fusion, Kalman filtering, obstacle avoidance, hardware/software integration, Git, GitHub, MATLAB